FOR SEPARATED MORPHISMS THE EXISTENCE PART OF THE VALUATIVE CRITERION MAY BE TESTED WITH SINGLE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author states that only (1) implies (2) has to be shown; the converse is the special case of (1) in which the extension of fields is the identity, and the proof then produces the morphism via the push-down lemma.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The two formulations of the valuative criterion. Definition definition-valuative-criterion and Schemes, Definition schemes-definition-valuative-criterion, used at proof lines 8618-8632. Fix the two conditions being compared and the two displayed diagrams.
- The push-down lemma for separated morphisms. Lemma lemma-push-down-solution, used at proof lines 8651-8653. Turns the K'/A' solution into the required morphism Spec(A) -> X.
- Separatedness of f is the hypothesis that makes the push-down available. used at proof lines 8617-8618. Is the standing assumption of the statement.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0ARH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13545–13545. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01KD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2046–2046. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a separated morphism of algebraic spaces over S. The following are equivalent*

(1) *f satisfies the existence part of the valuative criterion as in Definition 03IX,*

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(2) *given any commutative solid diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where A is a valuation ring with field of fractions K, there exists a dotted arrow, i.e., f satisfies the existence part of the valuative criterion as in Schemes, Definition 01KD.

Proof. We have to show that (1) implies (2). Suppose given a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

as in part (2). By (1) there exists a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & & \nearrow & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

as in Definition 03IX with $K \subset K'$ arbitrary. By Lemma 0ARH we can find a morphism $\mathrm{Spec}(A) \to X$ fitting into the diagram, i.e., (2) holds. □